Exercise 7.1

SOLUTION

Q.1. Evaluate each of the following:

i. $\frac{10!}{0!8!}$

Sol: $\frac{10.9.8!}{1.8!} = 90 \qquad (\because 0! = 1)$

ii. $\frac{12!}{3!(12-3)!}$

Sol: $\frac{12!}{3!.9!} = \frac{12.11.10.9!}{3.2.1.9!} = 2.11.10 = 220$

iii. $\frac{1440}{3!4!} + \frac{2400}{5!2!}$

Sol: $\frac{1440}{3!.4!} + \frac{2400}{5!2!} = \frac{1440}{3.2.1.4.3.2.1} + \frac{2400}{5.4.3.2.1.2.1}$

$= \frac{1440}{6\times 24} + \frac{2400}{120\times 2} = \frac{1440}{144} + \frac{2400}{240}$

$= 10 + 10 = 20$

iv. $\frac{(n+2)!}{(n+1)!}$

Sol: $\frac{(n+2)!}{(n+1)!} = \frac{(n+2)(n+1)!}{(n+1)!}$

$= \frac{(n+2)\,(n+1)!}{(n+1)!} = (n+2)$

Q.2. Write each of the following in the factorial form:

i. $n^3 - n$

Sol: $n^3 - n$

$n(n^2-1) = n(n+1)(n-1)$

$\frac{(n+1).n.(n-1).(n-2)!}{(n-2)!} = \frac{(n+1)!}{(n-2)!}$

ii. $n(n-1)(n-2).....(n-r+1)$

Sol: $\frac{n(n-1)(n-2).....(n-r+1)\times(n-r)!}{(n-r)!} = \frac{n!}{(n-r)!}$

Q.3. Find n, if $(n+4)! = 3024.n!$

Sol: $(n+4)! = 3024.n!$

$(n+4)(n+3)(n+2)(n+1)n! = 3024.n!$

Divide both sides by $n!$

$(n+4)(n+3)(n+2)(n+1) = 3024$

$(n+4)(n+3)(n+2)(n+1) = 9.8.7.6$

$\Rightarrow n+4 = 9 \Rightarrow \quad n = 5$

نوٹ:
Equation کو ڈائریکٹ حل کرنے کے لیے جیسا کہ Q.3 کے آخر میں کیا گیا ہے۔ دونوں سائیڈ پر 4 Factors ہیں اور ترتیب سے لکھے ہوئے ہیں۔ ہم نے دونوں طرف کے پہلے Factor کو برابر لیا ہے۔ تو n کی قیمت آجاتی ہے۔

Q.4. If $\frac{1}{7!} + \frac{1}{8!} = \frac{x}{9!}$, find x.

Sol: $\frac{1}{7!} + \frac{1}{8!} = \frac{x}{9!}$

$\frac{1}{7!} + \frac{1}{8.7!} = \frac{x}{9!}$

$\frac{1}{7!}\left(1 + \frac{1}{8}\right) = \frac{x}{9!}$

$\frac{1}{7!}\left(\frac{8+1}{8}\right) = \frac{x}{9!}$

$\frac{9}{8.7!} = \frac{x}{9!}$

To make $9!$ in denominator of L.H.S multiply and divide L.H.S by 9

$\frac{9}{7!.8} \times \frac{9}{9} = \frac{x}{9!}$

$\frac{81}{9.8.7!} = \frac{x}{9!}$

$\frac{81}{9!} = \frac{x}{9!} \Rightarrow \quad x = 81$

Q.5. Prove that $\frac{(2n+1)!}{n!} = [1.3.5...(2n-1)(2n+1)]2^n$

Sol: L.H.S =

$\frac{(2n+1)!}{n!} = \frac{(2n+1)(2n)(2n-1)....5.4.3.2.1}{n!}$

$= \frac{\{(2n+1)(2n-1)....5.3.1\}\{2n....6.4.2\}}{n!}$

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$$= \frac{2^n \{1.2.3.....n\}.\{1.3.5.....(2n-1)(2n+1)\}}{n!}$$

(Take 2 common from all even terms)

$$= \frac{2^n . n! \{1.3.5.....(2n-1)(2n+1)\}}{n!}$$

$$= 2^n \{1.3.5.....(2n-1)(2n+1)\}$$

$$= \{1.3.5.....(2n-1)(2n+1)\} 2^n = \text{R.H.S}$$

Q.6. **Express as a single fraction:** $\frac{(n+2)!}{(r+2)!} + \frac{(n+1)!}{(r+1)!}$

Sol:

$$\frac{(n+2)!}{(r+2)!} + \frac{(n+1)!}{(r+1)!}$$

$$\frac{(n+2)(n+1)!}{(r+2)(r+1)!} + \frac{(n+1)!}{(r+1)!} = \frac{(n+1)!}{(r+1)!}\left[\frac{n+2}{r+2} + 1\right]$$

$$= \frac{(n+1)!}{(r+1)!}\left[\frac{n+2+r+2}{r+2}\right]$$

$$= \frac{(n+1)! . (n+r+4)}{(r+2)(r+1)!}$$

$$= \frac{(n+1)!(n+r+4)}{(r+2)!}$$

Q.7. **There are four distinct colored balls and four boxes of same colors as those of the balls. Determine the number of possible ways the balls, one each in a box, can be placed such that a ball does not go to a box of its own color?**

Sol: Key concept arrangement of n things

$B_1 \longrightarrow C_1$

$B_2 \longrightarrow C_2$

$B_3 \longrightarrow C_3$

$B_4 \longrightarrow C_4$

$$D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!}.....n\right]$$

Arrangement of 4 balls in 4 boxes

$$= 4!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!}\right]$$

$$= 24\left[1 - 1 + \frac{1}{2} - \frac{1}{3.2.1} + \frac{1}{4.3.2.1}\right]$$

$$= 24\left[\frac{1}{2} - \frac{1}{6} + \frac{1}{24}\right] = 24\left[\frac{12-4+1}{24}\right]$$

$$= 24 \times \frac{9}{24} = 9 \text{ Ways}$$

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